3D Scene Artifact Narrative

Robert Rossman

Global Campus, Southern New Hampshire University

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Instructor Anna Sandifer

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Briefly describe the artifact. What is it? When was it created?

The artifact I enhanced for the software design and engineering category of my capstone is a 3D scene I created in SNHU's CS-330: Computer Graphics and Visualization course in December of 2025. It uses OpenGL to render a 3D scene built based on a series of reference photographs and features objects modeled using a combination of simple 3D shapes that are textured appropriately. The user can move the camera around the scene using the W, A, S, and D keys for horizontal movement, the Q and E keys for vertical movement, and the mouse to control the camera angle. The user can also switch between a perspective and orthographic display using the P and O keys.

Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

I selected this item as I believe it showcases several skills, such as 3D modeling, custom texturing, lighting, and implementing interactivity while considering usability. This artifact shows these skills applied in a practical setting, with each coming together to complete the scene.

The artifact was improved by adding the ability to display some of the reference photos used to build the scene. By pressing the R key, the user can now toggle a reference photo display, and by pressing the F key, the user can cycle to the next photo. When a reference photo is displayed, the scene's camera is positioned and angled to closely match the photo.

Additionally, the artifact was improved by adding the ability to change the lighting and background color of the scene. By pressing the L key, the user can now cycle through a series of

eight lighting presets. Similarly, by pressing the B key, the user can now cycle through a series of 14 preset background colors.

Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

Originally, I planned on meeting course outcomes two and four with this enhancement and I feel that I have met these outcomes. I think the artifact's value has improved significantly with the addition of the reference photo display, and lighting and background settings, which aligns it well with course outcome four. I also think the enhancements align with course outcome four as the reference photo display is communicated especially well with it being clear how the reference images shown relate to the 3D scene. On top of this, the reference photo display is shown in a separate window to avoid blocking the view of the 3D scene and increase usability.

However, there is some room for improvement. Most glaringly to me is the absence of a way to see the controls for the scene. After the enhancements, there are more than twelve different inputs the user can use to manipulate the scene. A toggleable display with the controls that is shown when opening the scene would help greatly with this.

Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

The process of enhancing and modifying the artifact was quite difficult but also rewarding. While I had experience in OpenGL from the class I built the scene in, most of what I was attempting to accomplish was new to me. Notably, I did not know how to create a UI element that stays on screen regardless of the camera's position as I had only worked with 3D objects. Additionally, while I already had the code to create and bind a texture, I did not know

how to apply this to a UI element. I had many smaller challenges too, such as needing to figure out I had to use forward declaration due to having two headers include one another.

My biggest diversion came from trying to display the reference photos in a separate window. I completed most of the code for this, even having a window scaled based on the main window and positioned directly beside it, but I struggled with drawing the photos on the second window. While polishing the artifact, I discovered that my issue involved neglecting to share resources from the main window when creating the new window. This troubleshooting process also helped me implement the lighting and background settings by increasing my familiarity with GLFW functions.